Lab 3



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Submitted to

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SUBJECT

OPERATING SYSTEM

B.S (5TH SEMESTER) in SOFTWARE ENGINEERING

Statement of Purpose:

This lab will introduce the Directory and File related commands to you. We will start with the some basic but important commands used to navigate through the Linux file system. Then we will discuss Directory and File related commands. Finally, we will introduce the I/O redirection in Linux

```
afzaalpc@ubuntu:~$ pwd
/home/afzaalpc
afzaalpc@ubuntu:~$ cd /etc
afzaalpc@ubuntu:/etc$ pwd
/etc
afzaalpc@ubuntu:/etc$ cd ..
afzaalpc@ubuntu:/$ pwd
/
afzaalpc@ubuntu:/$ cd /
afzaalpc@ubuntu:/$ pwd
/
afzaalpc@ubuntu:/$ ls
bin   cdrom  etc   labb  lib64       media  opt   root  sbin  srv  tmp  var
boot  dev    home  lib   lost+found  mnt    proc  run   snap  sys  usr
afzaalpc@ubuntu:/$ ls -l
total 76
lrwxrwxrwx   1 root root     7 Apr 20 08:46 bin -> usr/bin
drwxr-xr-x   3 root root  4096 Sep 18 11:53 boot
dr-xr-xr-x   2 root root  4096 Sep 18 11:24 cdrom
drwxr-xr-x  19 root root  4220 Oct  8 04:00 dev
drwxr-xr-x 142 root root 12288 Oct  1 04:18 etc
drwxr-xr-x   3 root root  4096 Sep 18 11:54 home
drwxr-xr-x   2 root root  4096 Oct  6 14:24 labb
lrwxrwxrwx   1 root root     7 Apr 20 08:46 lib -> usr/lib
lrwxrwxrwx   1 root root     9 Apr 20 08:46 lib64 -> usr/lib64
drwx------   2 root root 16384 Sep 18 11:28 lost+found
drwxr-xr-x   2 root root  4096 Aug 26 05:27 media
drwxr-xr-x   2 root root  4096 Aug 26 05:27 mnt
drwxr-xr-x   2 root root  4096 Aug 26 05:27 opt
dr-xr-xr-x 288 root root     0 Oct  8 03:59 proc
drwx------   5 root root  4096 Sep 18 12:11 root
drwxr-xr-x  42 root root   960 Oct  8 04:00 run
lrwxrwxrwx   1 root root     8 Apr 20 08:46 sbin -> usr/sbin
```

```
drwxr-xr-x  16 root root  4096 Aug 26 05:35 snap
drwxr-xr-x   2 root root  4096 Aug 26 05:27 srv
dr-xr-xr-x  13 root root     0 Oct  8 03:59 sys
drwxrwxrwt  15 root root   360 Oct  8 04:16 tmp
drwxr-xr-x  12 root root  4096 Aug 26 05:27 usr
drwxr-xr-x  14 root root  4096 Sep 18 12:10 var
afzaalpc@ubuntu:/$ ls /etc
ModemManager              debuginfod          init.d              opt          speech-
dispatcher
NetworkManager            default             initramfs-tools     os-release   ssh
PackageKit                deluser.conf        inputrc             pam.conf     ssl
UPower                    depmod.d            insserv.conf.d      pam.d        sssd
X11                       dhcp                ipp-usb             paperspecs   subgid
adduser.conf              dhcpcd.conf         issue               passwd       subgid-
alsa                      dictionaries-common issue.net           passwd-      subuid
alternatives              dpkg                kdump               pcmcia       subuid-
anacrontab                dracut.conf         kernel              perl         sudo.conf
apm                       dracut.conf.d       krb5.conf.d         pki          sudo_logsrvd.conf
apparmor                  dracut.conf.dpkg-new ld.so.cache        plymouth     sudoers
apparmor.d                e2scrub.conf        ld.so.conf          pm           sudoers.d
apport                    emacs               ld.so.conf.d        pnm2ppa.conf supercat
apt                       environment         ldap                polkit-1     sysctl.d
avahi                     environment.d       legal               ppp          sysstat
bash.bashrc               ethertypes          libao.conf          profile      systemd
bash_completion           fonts               libaudit.conf       profile.d    terminfo
bash_completion.d         fprintd.conf        libblockdev         protocols    thermald
bindresvport.blacklist    fstab               libibverbs.d        pulse        timezone
binfmt.d                  fuse.conf           libnl-3             python3      tmpfiles.d
bluetooth                 fwupd               locale.conf         python3.14   tmpjtd3ndbp
brlapi.key                gai.conf            locale.gen          rc0.d        tpm2-tss
brltty                    gdb                 localtime           rc1.d        ubuntu-advantage
brltty.conf               gdm3                logcheck            rc2.d        ucf.conf
ca-certificates           geoclue             login.defs          rc3.d        udev
ca-certificates.conf      ghostscript         logrotate.conf      rc4.d        udisks2
ca-certificates.conf.dpkg-old glvnd           logrotate.d         rc5.d        ufw
chatscripts               gnome               lsb-release         rc6.d        update-manager
chrony                    gnome-remote-desktop machine-id         rcS.d        update-
motd.d
cloud                     gnutls              magic               resolv.conf  update-notifier
colord                    gprofng.rc          magic.mime          rmt
usb_modeswitch.conf
console-setup             groff               manpath.config      rpc
```

usb_modeswitch.d
cracklib group mime.types rsyslog.conf vconsole.conf
credstore group- mke2fs.conf rsyslog.d vim
credstore.encrypted grub.d modprobe.d rygel.conf vtrgb
cron.d gshadow modules sane.d vulkan
cron.daily gshadow- modules-load.d screenrc wgetrc
cron.hourly gss mtab security whoopsie
cron.monthly gtk-3.0 nanorc selinux wpa_supplicant
cron.weekly hdparm.conf netconfig sensors.d xattr.conf
cron.yearly host.conf netplan sensors3.conf xdg
crontab hostname network services xemacs21
cups hosts networkd-dispatcher sgml xml
cupshelpers hosts.allow networks shadow
zsh_command_not_found
dbus-1 hosts.deny newt shadow-
dconf hp nftables.conf shells
debconf.conf ifplugd nsswitch.conf skel
debian_version init openvpn snmp
afzaalpc@ubuntu:/$ ls /bin /sbin
/bin:
3cpio gpu-manager psicc
GET grdctl pslog
HEAD grep pstree
POST gresource pstree.x11
X11 groff ptar
Xwayland grog ptardiff
'[' grops ptargrep
aa-enabled grotty ptx
aa-exec groups ptyxis
aa-features-abi growpart pw-cat
aconnect grub-editenv pw-cli
acpidbg grub-file pw-config
add-apt-repository grub-fstest pw-container
addr2line grub-glue-efi pw-dot
airscan-discover grub-kbdcomp pw-dsdplay
alsabat grub-menulst2cfg pw-dump
alsaloop grub-mkfont pw-encplay
alsamixer grub-mkimage pw-link
alsatplg grub-mklayout pw-loopback
alsaucm grub-mknetdir pw-metadata
amidi grub-mkpasswd-pbkdf2 pw-midi2play
amixer grub-mkrelpath pw-midi2record

aplay
aplaymidi
aplaymidi2
apport-bug
apport-cli
apport-collect
apport-unpack
appres
appstreamcli
apropos
apt
apt-add-repository
apt-cache
apt-cdrom
apt-config
apt-get
apt-mark
aptdcon
ar
arch
arecord
arecordmidi
arecordmidi2
arm2hpdl
as
aseqdump
aseqnet
aseqsend
aspell
aspell-import
awk
axfer
b2sum
baobab
base32
base64
basename
basenc
bash
bashbug
bc
bdftopcf
grub-mkrescue
grub-mkstandalone
grub-mount
grub-ntldr-img
grub-render-label
grub-script-check
grub-syslinux2cfg
gs
gsbj
gsdj
gsdj500
gsettings
gslj
gslp
gsnd
gst-audio-thumbnailer
gst-device-monitor-1.0
gst-discoverer-1.0
gst-inspect-1.0
gst-launch-1.0
gst-play-1.0
gst-stats-1.0
gst-tester-1.0
gst-typefind-1.0
gst-video-thumbnailer
gstack
gstreamer-codec-install
gtbl
gted
gtk-builder-tool
gtk-encode-symbolic-svg
gtk-launch
gtk-query-settings
gtk-update-icon-cache
gtk4-broadwayd
gtk4-builder-tool
gtk4-encode-symbolic-svg
gtk4-image-tool
gtk4-launch
gtk4-path-tool
gtk4-query-settings
gtk4-rendernode-tool
pw-mididump
pw-midiplay
pw-midirecord
pw-mon
pw-play
pw-profiler
pw-record
pw-reserve
pw-sysex
pw-top
pwd
pwdx
py3clean
py3compile
py3versions
pybabel
pybabel-python3
pydoc3
pydoc3.14
pygettext3
pygettext3.14
pygmentize
pyserial-miniterm
pyserial-ports
python3
python3.14
pzstd
qmi-firmware-update
qmi-network
qmicli
qpdldecode
quirks-handler
ranlib
rbash
rctest
rdma
readelf
readlink
realpath
red
renice
reset

bdftruncate
bluemoon
bluetooth-sendto
bluetoothctl
boltctl
bpftrace
bpftrace-aotrt
brltty
brltty-atb
brltty-clip
brltty-ctb
brltty-hid
brltty-ktb
brltty-lscmds
brltty-morse
brltty-trtxt
brltty-ttb
brltty-tune
broadwayd
browse
btattach
btmgmt
btmon
busctl
busybox
bwrap
c++filt
c_rehash
calibrate_ppa
canberra-gtk-play
cancel
captoinfo
cat
catman
cd-create-profile
cd-fix-profile
cd-iccdump
cd-it8
chacl
chage
chardet
chardetect
gtk4-update-icon-cache
gunzip
gzexe
gzip
h2ph
h2xs
hardlink
hashsum
hbpldecode
hciattach
hciconfig
hcitool
hd
head
heif-thumbnailer
helpztags
hex2hcd
hexdump
hipercdecode
host
hostid
hostname
hostnamectl
hp-align
hp-check
hp-clean
hp-colorcal
hp-config_usb_printer
hp-doctor
hp-firmware
hp-info
hp-levels
hp-logcapture
hp-makeuri
hp-pkservice
hp-plugin
hp-plugin-ubuntu
hp-probe
hp-query
hp-scan
hp-setup
hp-testpage
resizecons
resolvectl
resources
rev
rfcomm
rgrep
rm
rmdir
rnano
routel
rpcgen
rrsync
rsync
rsync-ssl
rtla
rtstat
run-parts
run-with-aspell
run0
runcon
rview
rygel
sadf
sane-find-scanner
sar
sar.sysstat
savelog
sbattach
sbkeysync
sbsiglist
sbsign
sbvarsign
sbverify
scanimage
scp
scp-dbus-service
screen
screendump
script
scriptlive
scriptreplay
sdiff

chattr
chcon
check-language-support
chfn
chgrp
chmod
choom
chown
chronyc
chroot
chrt
chsh
chvt
cifsiostat
ciptool
ckbcomp
cksum
clear
clear_console
cloud-id
cloud-init
cloud-init-per
cmp
codepage
col
colcrt
colormgr
colrm
column
comm
corelist
coreutils
cp
cpan
cpan5.40-x86_64-linux-gnu
cpio
cpp
cpp-15
cpupower
crash
crontab
csplit
hp-timedate
hwctl
hwe-support-status
i386
ibus
ibus-daemon
ibus-setup
ibus-table-createdb
iceauth
iconv
id
iecset
ijs_pxljr
im-config
im-launch
inetutils-telnet
info
infobrowser
infocmp
infotocap
inputattach
install
install-info
instmodsh
ionice
iostat
ip
ipcmk
ipcrm
ipcs
ippfind
ipptool
iptables-xml
ischroot
ispell-wrapper
join
journalctl
jpgicc
jq
json-patch-jsondiff
json_pp
jsondiff
sdptool
seahorse
sed
select-default-iwrap
select-editor
sensible-browser
sensible-editor
sensible-pager
sensible-terminal
seq
sessreg
setarch
setfacl
setfont
setkeycodes
setleds
setlogcons
setmetamode
setpci
setpriv
setsid
setterm
setupcon
setxkbmap
sftp
sh
sha1sum
sha224sum
sha256sum
sha3-224sum
sha3-256sum
sha3-384sum
sha3-512sum
sha384sum
sha3sum
sha512sum
shake128sum
shake256sum
shasum
showconsolefont
showkey
showrgb

ctstat
cupstestppd
cut
cvtsudoers.ws
dash
date
dbus-cleanup-sockets
dbus-daemon
dbus-monitor
dbus-run-session
dbus-send
dbus-update-activation-environment
dbus-uuidgen
dbxtool
dc
dconf
dd
ddstdecode
deallocvt
deb-systemd-helper
deb-systemd-invoke
debconf
debconf-apt-progress
debconf-communicate
debconf-copydb
debconf-escape
debconf-set-selections
debconf-show
debian-distro-info
delv
desktop-file-edit
desktop-file-install
desktop-file-validate
df
dh_bash-completion
dh_installxmlcatalogs
dh_perl_openssl
diff
diff3
dig
dir
dircolors
jsonpatch
jsonpointer
jsonschema
kbd_mode
kbdinfo
kbxutil
kernel-install
kill
killall
kmod
kmodsign
l2ping
l2test
laptop-detect
lavadecode
ld
ld.bfd
ld.so
ldapadd
ldapcompare
ldapdelete
ldapexop
ldapmodify
ldapmodrdn
ldappasswd
ldapsearch
ldapurl
ldapwhoami
ldd
less
lessecho
lessfile
lesskey
lesspipe
lexgrog
libnetcfg
link
linkicc
linux-boot-prober
linux-check-removal
linux-run-hooks
linux-update-symlinks
shred
shuf
size
skill
slabtop
sleep
slxdecode
snap
snapctl
snapfuse
snice
soelim
sort
spa-acp-tool
spa-inspect
spa-json-dump
spa-monitor
spa-resample
spd-conf
spd-say
spdsend
speaker-test
speech-dispatcher
spice-vdagent
splain
split
splitfont
sqfscat
sqfstar
ss
ssh
ssh-add
ssh-agent
ssh-argv0
ssh-copy-id
ssh-keygen
ssh-keyscan
sss_ssh_authorizedkeys
sss_ssh_knownhosts
sss_ssh_knownhostsproxy
stat
static-sh

dirmngr
dirmngr-client
dirname
distro-info
dmesg
dnsdomainname
do-release-upgrade
domainname
dpkg
dpkg-deb
dpkg-divert
dpkg-maintscript-helper
dpkg-query
dpkg-realpath
dpkg-split
dpkg-statoverride
dpkg-trigger
dracut
dracut-catimages
driverless
driverless-fax
du
dumpkeys
dvipdf
eatmydata
ec2metadata
echo
ed
editor
editres
egrep
eject
elfedit
enc2xs
encguess
enchant-2
enchant-lsmod-2
env
envsubst
eps2eps
eqn
esc-m
linux-version
linux32
linux64
listres
ln
lnstat
loadkeys
loadunimap
locale
locale-check
localectl
localedef
localsearch
logger
login
loginctl
logname
look
loupe
lowntfs-3g
lp
lpoptions
lpq
lpr
lprm
lpstat
ls
lsattr
lsb_release
lsblk
lscpu
lshw
lsinitramfs
lsinitrd
lsipc
lslocks
lsmod
lsns
lsof
lspci
lspgpot
lspower
stdbuf
strace
strace-log-merge
streamzip
strings
strip
stty
su
su-rs
sudo
sudo-rs
sudo.ws
sudoedit
sudoedit-rs
sudoreplay.ws
sum
switcherooctl
sync
sysprof
sysprof-agent
sysprof-cat
sysprof-cli
systemctl
systemd-ac-power
systemd-analyze
systemd-ask-password
systemd-cat
systemd-cgls
systemd-cgtop
systemd-confext
systemd-creds
systemd-cryptenroll
systemd-cryptsetup
systemd-delta
systemd-detect-virt
systemd-escape
systemd-firstboot
systemd-hwdb
systemd-id128
systemd-inhibit
systemd-machine-id-setup
systemd-mount

eutp
ex
expand
expiry
expr
factor
fallocate
false
fc-cache
fc-cat
fc-conflist
fc-list
fc-match
fc-pattern
fc-query
fc-scan
fc-validate
fgconsole
fgrep
file
file2brl
find
findmnt
firefox
flock
fmt
fold
fonttosfnt
foo2ddst
foo2ddst-wrapper
foo2hbpl2
foo2hbpl2-wrapper
foo2hiperc
foo2hiperc-wrapper
foo2hp
foo2hp2600-wrapper
foo2lava
foo2lava-wrapper
foo2oak
foo2oak-wrapper
foo2qpdl
foo2qpdl-wrapper
lsusb
luit
lwp-download
lwp-dump
lwp-mirror
lwp-request
lzcat
lzcmp
lzdiff
lzegrep
lzfgrep
lzgrep
lzless
lzma
lzmainfo
lzmore
m17n-db
m2300w
m2300w-wrapper
m2400w
makedumpfile
makedumpfile-R.pl
man
man-recode
mandb
manpath
mapscrn
markdown-it
mawk
mbim-network
mbimcli
mcookie
md5sum
mdig
memhog
mesa-overlay-control.py
mesa-screenshot-control.py
migrate-pubring-from-classic-gpg
migratepages
migspeed
mimeopen
mimetype
systemd-mute-console
systemd-notify
systemd-path
systemd-pty-forward
systemd-run
systemd-socket-activate
systemd-stdio-bridge
systemd-sysext
systemd-sysusers
systemd-tmpfiles
systemd-tty-ask-password-agent
systemd-umount
systemd-vpick
tabs
tac
tail
tapestat
tar
taskset
tbl
tclsh
tclsh8.6
tcpdump
tecla
tee
telnet
tempfile
test
tic
tificc
time
timedatectl
timeout
tinysparql
tload
tnftp
toe
top
touch
tput
tr
trace-cmd

foo2slx
foo2slx-wrapper
foo2xqx
foo2xqx-wrapper
foo2zjs
foo2zjs-icc2ps
foo2zjs-pstops
foo2zjs-wrapper
foomatic-rip
fprintd-delete
fprintd-enroll
fprintd-list
fprintd-verify
free
ftp
funzip
fuser
fusermount
fusermount3
fwupdmgr
fwupdtool
gamemoded
gamma4scanimage
gapplication
gatttool
gcalccmd
gcore
gcr-viewer
gcr-viewer-gtk4
gdb
gdb-add-index
gdbtui
gdbus
gdctl
gdk-pixbuf-csource
gdk-pixbuf-pixdata
gdm-config
gencat
geqn
getconf
getent
getfacl
min12xxw
mk_modmap
mkdir
mkfifo
mkfontdir
mkfontscale
mknod
mksquashfs
mktemp
mmcli
mokutil
monitor-sensor
more
mount
mountpoint
mpris-proxy
mpstat
mscompress
msexpand
mt
mt-gnu
mtr
mtr-packet
mv
namei
nano
nautilus
nautilus-autorun-software
nautilus-sendto
nawk
nc
nc.openbsd
neqn
netaddr
netcat
netshaper
networkctl
networkd-dispatcher
ngettext
nhlt-dmic-info
nice
nisdomainname
tracepath
transicc
troff
true
truncate
trust
tset
tsort
tty
turbostat
tzselect
ua
ubuntu-advantage
ubuntu-bug
ubuntu-distro-info
ubuntu-drivers
ubuntu-insights
ubuntu-report
ubuntu-security-status
ucf
ucfq
ucfr
uclampset
ucs2any
udevadm
udisksctl
ul
umax_pp
umount
uname
unattended-upgrade
unattended-upgrades
uncompress
unexpand
unicode_start
unicode_stop
uniq
unity-scope-loader
unlink
unlzma
unmkinitramfs
unshare

getkeycodes
getopt
gettext
gettext.sh
ghostscript
ginstall-info
gio
gio-querymodules
gipddecode
gjs
gjs-console
glib-compile-schemas
glycin-thumbnailer
gnome-calculator
gnome-characters
gnome-clocks
gnome-control-center
gnome-disk-image-mounter
gnome-disks
gnome-extensions
gnome-font-viewer
gnome-help
gnome-keyring
gnome-keyring-3
gnome-keyring-daemon
gnome-language-selector
gnome-logs
gnome-service-client
gnome-session
gnome-session-inhibit
gnome-session-quit
gnome-shell
gnome-shell-extension-tool
gnome-shell-test-tool
gnome-text-editor
gnome-thumbnail-font
gnome-www-browser
'gnu['
gnuarch
gnub2sum
gnubase32
gnubase64
nl
nm
nm-connection-editor
nm-online
nmcli
nmtui
nmtui-connect
nmtui-edit
nmtui-hostname
nohup
notify-send
nproc
nroff
nsenter
nslookup
nstat
nsupdate
ntfs-3g
ntfs-3g.probe
ntfscat
ntfscluster
ntfscmp
ntfsdecrypt
ntfsfallocate
ntfsfix
ntfsinfo
ntfsls
ntfsmove
ntfsrecover
ntfssecaudit
ntfstruncate
ntfsusermap
ntfswipe
numactl
numastat
numfmt
oakdecode
obexctl
objcopy
objdump
od
oem-getlogs
unsquashfs
unxz
unzip
unzipsfx
unzstd
update-alternatives
update-desktop-database
update-manager
update-mime-database
update-notifier
upower
uptime
usb-devices
usb_printerid
usbhid-dump
usbip
usbipd
usbreset
user-session-migration
users
uuidgen
uuidparse
varlinkctl
vcs-run
vdir
vi
view
viewres
vim.tiny
visudo-rs
vmstat
vstp
w
wall
watch
watchgnupg
wc
wget
whatis
whereis
which
which.debianutils

gnubasename
gnubasenc
gnucat
gnuchcon
gnuchgrp
gnuchmod
gnuchown
gnucksum
gnucomm
gnucp
gnucsplit
gnucut
gnudate
gnudd
gnudf
gnudir
gnudircolors
gnudirname
gnudu
gnuecho
gnuenv
gnuexpand
gnuexpr
gnufactor
gnufalse
gnufmt
gnufold
gnugroups
gnuhead
gnuhostid
gnuid
gnuinstall
gnujoin
gnulink
gnuln
gnulogname
gnuls
gnumd5sum
gnumkdir
gnumkfifo
gnumknod
gnumktemp
on_ac_power
oomctl
open
openssl
openvt
opldecode
orca
orca-dm-wrapper
os-prober
p11-kit
pager
paper
paperconf
papers
papers-previewer
papers-thumbnailer
partx
passwd
paste
patch
pathchk
pcilmr
pdb3
pdb3.14
pdf2dsc
pdf2ps
pdfattach
pdfdetach
pdffonts
pdfimages
pdfinfo
pdfseparate
pdfsig
pdftocairo
pdftohtml
pdftoppm
pdftops
pdftotext
pdfunite
peekfd
perf
perl
whiptail
who
whoami
whoopsie
whoopsie-preferences
wireplumber
word-list-compress
wpa_passphrase
wpctl
wpexec
wsdd
x-session-manager
x-terminal-emulator
x-www-browser
x86_64
x86_64-linux-gnu-addr2line
x86_64-linux-gnu-ar
x86_64-linux-gnu-as
x86_64-linux-gnu-c++filt
x86_64-linux-gnu-cpp
x86_64-linux-gnu-cpp-15
x86_64-linux-gnu-elfedit
x86_64-linux-gnu-gprof
x86_64-linux-gnu-ld
x86_64-linux-gnu-ld.bfd
x86_64-linux-gnu-nm
x86_64-linux-gnu-objcopy
x86_64-linux-gnu-objdump
x86_64-linux-gnu-ranlib
x86_64-linux-gnu-readelf
x86_64-linux-gnu-size
x86_64-linux-gnu-strings
x86_64-linux-gnu-strip
x86_energy_perf_policy
xargs
xauth
xbrlapi
xcmsdb
xdg-dbus-proxy
xdg-desktop-icon
xdg-desktop-menu
xdg-email

gnumv
gnunice
gnunl
gnunohup
gnunproc
gnunumfmt
gnuod
gnupaste
gnupathchk
gnupinky
gnupr
gnuprintenv
gnuprintf
gnuptx
gnupwd
gnureadlink
gnurealpath
gnurm
gnurmdir
gnuruncon
gnuseq
gnusha1sum
gnusha224sum
gnusha256sum
gnusha384sum
gnusha512sum
gnushred
gnushuf
gnusleep
gnusort
gnusplit
gnustat
gnustdbuf
gnustty
gnusum
gnusync
gnutac
gnutail
gnutee
gnutest
gnutimeout
gnutouch
perl5.40-x86_64-linux-gnu
perl5.40.1
perlbug
perldoc
perli11ndoc
perlivp
perlthanks
pf2afm
pfbtopfa
pgrep
pic
pico
piconv
pidof
pidstat
pidwait
pinentry
pinentry-curses
pinentry-gnome3
pinentry-x11
ping
ping4
ping6
pinky
pipewire
pipewire-aes67
pipewire-avb
pipewire-pulse
pkaction
pkcheck
pkexec
pkgcli
pkill
pkttyagent
pl2pm
pldd
plog
plymouth
pmap
pnm2ppa
pod2html
pod2man
xdg-icon-resource
xdg-mime
xdg-open
xdg-screensaver
xdg-settings
xdg-terminal-exec
xdg-user-dir
xdg-user-dirs-gtk-update
xdg-user-dirs-update
xdpyinfo
xdriinfo
xev
xfd
xfontsel
xgamma
xhost
xkbbell
xkbcomp
xkbevd
xkbprint
xkbvleds
xkbwatch
xkeystone
xkill
xlsatoms
xlsclients
xlsfonts
xmessage
xmodmap
xprop
xqxdecode
xrandr
xrdb
xrefresh
xset
xsetmode
xsetpointer
xsetroot
xstdcmap
xsubpp
xvidtune
xvinfo

gnutr
gnutrue
gnutruncate
gnutsort
gnutty
gnuuname
gnuunexpand
gnuuniq
gnuunlink
gnuusers
gnuvdir
gnuwc
gnuwho
gnuwhoami
gnuyes
gp-archive
gp-collect-app
gp-display-html
gp-display-src
gp-display-text
gpasswd
gpg
gpg-agent
gpg-connect-agent
gpg-mail-tube
gpg-wks-client
gpgconf
gpgparsemail
gpgsm
gpgsplit
gpgtar
gpgv
gpic
gprof
gprofng
gprofng-archive
gprofng-collect-app
gprofng-display-html
gprofng-display-src
gprofng-display-text
gprofng-gmon
pod2text
pod2usage
podchecker
poff
pon
powerprofilesctl
ppdc
ppdhtml
ppdi
ppdmerge
ppdpo
pphs
pr
precat
preconv
preunzip
prezip
prezip-bin
printafm
printenv
printer-profile
printf
prlimit
pro
prove
prtstat
ps
ps2ascii
ps2epsi
ps2pdf
ps2pdf12
ps2pdf13
ps2pdf14
ps2pdfwr
ps2ps
ps2ps2
ps2txt
psfaddtable
psfgettable
psfstriptable
psfxtable
xwininfo
xxd
xz
xzcat
xzcmp
xzdiff
xzegrep
xzfgrep
xzgrep
xzless
xzmore
yelp
yes
ypdomainname
zcat
zcmp
zdiff
zdump
zegrep
zenity
zfgrep
zforce
zgrep
zip
zipcloak
zipdetails
zipgrep
zipinfo
zipnote
zipsplit
zjsdecode
zless
zmore
znew
zstd
zstdcat
zstdgrep
zstdless
zstdmt

/sbin:
ModemManager ethtool mkdosfs softirqs-bpfcc
NetworkManager execsnoop-bpfcc mke2fs solisten-bpfcc
aa-load execsnoop.bt mkfs spice-vdagentd
aa-remove-unknown exfat2img mkfs.exfat ssllatency.bt
aa-status exfatlabel mkfs.ext2 sslsniff-bpfcc
aa-teardown exitsnoop-bpfcc mkfs.ext3 sslsnoop.bt
accessdb ext4dist-bpfcc mkfs.ext4 sssd
add-shell ext4slower-bpfcc mkfs.fat stackcount-bpfcc
addgnupghome f2fsslower-bpfcc mkfs.msdos start-stop-daemon
addgroup faillock mkfs.ntfs statsnoop-bpfcc
adduser fatlabel mkfs.vfat statsnoop.bt
agetty fdisk mkhomedir_helper sudo_logsrvd
alsa filefrag mkinitramfs sudo_sendlog.ws
alsa-info filegone-bpfcc mklost+found sulogin
alsabat-test filelife-bpfcc mkntfs swapin.bt
alsactl fileslower-bpfcc mkswap swaplabel
anacron filetop-bpfcc modinfo swapoff
apparmor_parser findfs modprobe swapon
apparmor_status fixparts mount.fuse syncsnoop-bpfcc
applygnupgdefaults fsck mount.fuse3 syncsnoop.bt
aptd fsck.exfat mount.lowntfs-3g syscount-bpfcc
argdist-bpfcc fsck.ext2 mount.ntfs syscount.bt
arpd fsck.ext3 mount.ntfs-3g sysctl
arptables fsck.ext4 mountsnoop-bpfcc tarcat
arptables-nft fsck.fat mptcpify-bpfcc tc
arptables-nft-restore fsck.msdos mysqld_qslower-bpfcc tclcalls-bpfcc
arptables-nft-save fsck.vfat naptime.bt tclflow-bpfcc
arptables-restore fsfreeze netplan tclobjnew-bpfcc
arptables-save fstab-decode netqtop-bpfcc tclstat-bpfcc
arptables-translate fstrim newusers tcpaccept-bpfcc
aspell-autobuildhash funccount-bpfcc nfnl_osf tcpaccept.bt
avahi-daemon funcinterval-bpfcc nfsdist-bpfcc tcpcong-bpfcc
badblocks funclatency-bpfcc nfsslower-bpfcc tcpconnect-bpfcc
bashreadline-bpfcc funcslower-bpfcc nft tcpconnect.bt
bashreadline.bt gdisk nodegc-bpfcc tcpconnlat-bpfcc
bindsnoop-bpfcc gdm3 nodestat-bpfcc tcpdrop-bpfcc
biolatency-bpfcc genl nologin tcpdrop.bt
biolatency-kp.bt getcap ntfsclone tcplife-bpfcc
biolatency.bt gethostlatency-bpfcc ntfscp tcplife.bt
biolatpcts-bpfcc gethostlatency.bt ntfslabel tcpretrans-bpfcc
biopattern-bpfcc getpcaps ntfsresize tcpretrans.bt

biosdecode getty ntfsundelete tcprtt-bpfcc
biosnoop-bpfcc getweb numasched-bpfcc tcpstates-bpfcc
biosnoop.bt gnome-menus-blacklist offcputime-bpfcc tcpsubnet-bpfcc
biostacks.bt gnuchroot offwaketime-bpfcc tcpsynbl-bpfcc
biotop-bpfcc groupadd on_ac_power tcpsynbl.bt
bitesize-bpfcc groupdel oomkill-bpfcc tcptop-bpfcc
bitesize.bt groupmod oomkill.bt tcptracer-bpfcc
blkdeactivate grpck opensnoop-bpfcc thermald
blkdiscard grpconv opensnoop.bt threadsnoop-bpfcc
blkid grpunconv openvpn threadsnoop.bt
blockdev grub-bios-setup ownership tipc
bluetoothd grub-install pam-auth-update tplist-bpfcc
bpflist-bpfcc grub-macbless pam_extrausers_chkpwd trace-bpfcc
bpftool grub-mkconfig pam_extrausers_update ttysnoop-bpfcc
bridge grub-mkdevicemap pam_getenv tune.exfat
brltty-setup grub-probe pam_namespace_helper tune2fs
btrfsdist-bpfcc grub-reboot pam_timestamp_check u-d-c-print-pci-ids
btrfsslower-bpfcc grub-set-default paperconfig ucalls
cachestat-bpfcc halt parted uflow
cachetop-bpfcc hardirqs-bpfcc partprobe ufw
capable-bpfcc hdparm pccardctl ugc
capable.bt iconvconfig perlcalls-bpfcc umount.udisks2
capsh init perlflow-bpfcc undump.bt
cfdisk inject-bpfcc perlstat-bpfcc unix_chkpwd
cgdisk insmod phpcalls-bpfcc unix_update
chat install-sgmlcatalog phpflow-bpfcc uobjnew
chgpasswd installkernel phpstat-bpfcc update-ca-certificates
chpasswd invoke-rc.d pidpersec-bpfcc update-catalog
chronyd ip pidpersec.bt update-cracklib
cobjnew-bpfcc ip6tables plymouthd update-default-aspell
coldreboot ip6tables-apply policy-rc.d update-default-ispell
compactsnoop-bpfcc ip6tables-legacy poweroff update-default-wordlist
cpudist-bpfcc ip6tables-legacy-restore ppchcalls-bpfcc update-dictcommon-aspell
cpuunclaimed-bpfcc ip6tables-legacy-save pppd update-dictcommon-hunspell
cpuwalk.bt ip6tables-nft pppdump update-fonts-alias
cracklib-check ip6tables-nft-restore pppoe-discovery update-fonts-dir
cracklib-format ip6tables-nft-save pppstats update-fonts-scale
cracklib-packer ip6tables-restore pptp update-grub
cracklib-unpacker ip6tables-restore-translate pptpsetup update-grub-

gfxpayload
create-cracklib-dict ip6tables-save profile-bpfcc update-grub2
criticalstat-bpfcc ip6tables-translate pwck update-gsfontmap
cron ipp-usb pwconv update-icon-caches
cups-browsed ippevepcl pwhistory_helper update-ieee-data
cupsaccept ippeveprinter pwunconv update-inetd
cupsctl ippeveps pythoncalls-bpfcc update-info-dir
cupsd iptables pythonflow-bpfcc update-initramfs
cupsdisable iptables-apply pythongc-bpfcc update-locale
cupsenable iptables-legacy pythonstat-bpfcc update-passwd
cupsfilter iptables-legacy-restore rdmaucma-bpfcc update-pciids
cupsreject iptables-legacy-save readahead-bpfcc update-rc.d
dbslower-bpfcc iptables-nft readprofile update-shells
dbstat-bpfcc iptables-nft-restore reboot update-xmlcatalog
dcb iptables-nft-save remove-default-ispell usb_modeswitch
dcsnoop-bpfcc iptables-restore remove-default-wordlist
usb_modeswitch_dispatcher
dcsnoop.bt iptables-restore-translate remove-shell usbmuxd
dcstat-bpfcc iptables-save reset-trace-bpfcc useradd
deadlock-bpfcc iptables-translate resize2fs userdel
debugfs ispell-autobuildhash resolvconf usermod
delgroup iucode-tool rfkill ustat
deluser iucode_tool rmmod uthreads
depmod iw rmt uuidd
devlink javacalls-bpfcc rmt-tar validlocale
dhcpcd javaflow-bpfcc rsyslogd vcstime
dirtop-bpfcc javagc-bpfcc rtacct vdpa
dmidecode javaobjnew-bpfcc rtcwake vfscount-bpfcc
dmsetup javastat-bpfcc rtkitctl vfscount.bt
dmstats javathreads-bpfcc rtmon vfsstat-bpfcc
dnsmasq kbdrate rubycalls-bpfcc vfsstat.bt
dosfsck kdump-config rubyflow-bpfcc vigr
dosfslabel kexec rubygc-bpfcc vipw
dpkg-preconfigure kexec-load-kernel rubyobjnew-bpfcc virtiostat-bpfcc
dpkg-reconfigure killall5 rubystat-bpfcc visudo
dpll killsnoop-bpfcc runqlat-bpfcc visudo.ws
drsnoop-bpfcc killsnoop.bt runqlat.bt vmcore-dmesg
dump.exfat klockstat-bpfcc runqlen-bpfcc vpddecode
dumpe2fs kvmexit-bpfcc runqlen.bt wakeuptime-bpfcc
e2freefrag ldconfig runqslower-bpfcc wipefs
e2fsck llcstat-bpfcc runuser wpa_action
e2image loads.bt saned wpa_cli

```
e2label              locale-gen              select-default-ispell    wpa_supplicant
e2mmpstatus          logrotate               select-default-wordlist  wqlat-bpfcc
e2scrub              logsave                 service                  writeback.bt
e2scrub_all          losetup                 setcap                   xfsdist-bpfcc
e2undo               lpadmin                 setuids.bt               xfsdist.bt
e4crypt              lpc                     setvesablank             xfsslower-bpfcc
e4defrag             lpinfo                  setvtrgb                 xtables-legacy-multi
ebtables             lpmove                  sfdisk                   xtables-monitor
ebtables-nft         lsmod                   sgdisk                   xtables-nft-multi
ebtables-nft-restore lspcmcia                shadowconfig             zfsdist-bpfcc
ebtables-nft-save    make-ssl-cert           shmsnoop-bpfcc           zfsslower-bpfcc
ebtables-restore     mdflush-bpfcc           shutdown                 zic
ebtables-save        mdflush.bt              slabratetop-bpfcc        zramctl
ebtables-translate   memleak-bpfcc           sofdsnoop-bpfcc
afzaalpc@ubuntu:/$ ls
bin   cdrom  etc   labb  lib64       media  opt   root  sbin  srv  tmp  var
boot  dev    home  lib   lost+found  mnt    proc  run   snap  sys  usr
afzaalpc@ubuntu:/$ cd
afzaalpc@ubuntu:~$ ls -a
.   .bash_history  .bashrc  .config  .profile  Desktop    Downloads  Pictures  Templates
lab3
..  .bash_logout   .cache   .local   .ssh      Documents  Music      Public    Videos    snap
afzaalpc@ubuntu:~$ ls -lh /boot
total 137M
-rw------- 1 root root  11M Jul 31 16:27 System.map-7.0.0-30-generic
-rw------- 1 root root  11M Aug  1 02:09 System.map-7.0.0-31-generic
-rw-r--r-- 1 root root 302K Jul 31 16:27 config-7.0.0-30-generic
-rw-r--r-- 1 root root 302K Aug  1 02:09 config-7.0.0-31-generic
drwxr-xr-x 5 root root 4.0K Sep 18 11:54 grub
lrwxrwxrwx 1 root root   27 Sep 18 11:49 initrd.img -> initrd.img-7.0.0-31-generic
-rw------- 1 root root  41M Sep 18 11:53 initrd.img-7.0.0-30-generic
-rw------- 1 root root  41M Sep 18 11:50 initrd.img-7.0.0-31-generic
lrwxrwxrwx 1 root root   27 Sep 18 11:51 initrd.img.old -> initrd.img-7.0.0-30-generic
-rw-r--r-- 1 root root 153K Nov 29  2025 mt86+ia32
-rw-r--r-- 1 root root 158K Nov 29  2025 mt86+x64
lrwxrwxrwx 1 root root   24 Sep 18 11:49 vmlinuz -> vmlinuz-7.0.0-31-generic
-rw------- 1 root root  17M Jul 31 16:27 vmlinuz-7.0.0-30-generic
-rw------- 1 root root  17M Aug  1 02:01 vmlinuz-7.0.0-31-generic
lrwxrwxrwx 1 root root   24 Sep 18 11:49 vmlinuz.old -> vmlinuz-7.0.0-30-generic
afzaalpc@ubuntu:~$ cd
afzaalpc@ubuntu:~$ mkdir lab03
afzaalpc@ubuntu:~$ ls
```

Desktop Documents Downloads Music Pictures Public Templates Videos afzaal lab03 lab3 snap
afzaalpc@ubuntu:~$ mkdir f1 f2 f3
afzaalpc@ubuntu:~$ ls
Desktop Documents Downloads Music Pictures Public Templates Videos afzaal f1 f2 f3 lab03 lab3 snap
afzaalpc@ubuntu:~$ cd lab03
afzaalpc@ubuntu:~/lab03$ pwd
/home/afzaalpc/lab03
afzaalpc@ubuntu:~/lab03$ touch file1.txt
afzaalpc@ubuntu:~/lab03$ ls
file1.txt
afzaalpc@ubuntu:~/lab03$ touch file2.txt
afzaalpc@ubuntu:~/lab03$ ls
file1.txt file2.txt
afzaalpc@ubuntu:~/lab03$ cat > file1.txt
this is lab03 and myself afzaal
afzaalpc@ubuntu:~/lab03$ cat file1.txt
this is lab03 and myself afzaal
afzaalpc@ubuntu:~/lab03$ cat >> file1.txt
this is an os lab
afzaalpc@ubuntu:~/lab03$ cat file1.txt
this is lab03 and myself afzaal
this is an os lab
afzaalpc@ubuntu:~/lab03$ cp file1.txt file3.txt
afzaalpc@ubuntu:~/lab03$ ls
file1.txt file2.txt file3.txt
afzaalpc@ubuntu:~/lab03$ mv file2.txt newfile.txt
afzaalpc@ubuntu:~/lab03$ ls
file1.txt file3.txt newfile.txt
afzaalpc@ubuntu:~/lab03$ mkdir backup
afzaalpc@ubuntu:~/lab03$ mv file3.txt backup/
afzaalpc@ubuntu:~/lab03$ ls
backup file1.txt newfile.txt
afzaalpc@ubuntu:~/lab03$ ls backup
file3.txt
afzaalpc@ubuntu:~/lab03$ rm newfile.txt
afzaalpc@ubuntu:~/lab03$ ls
backup file1.txt
afzaalpc@ubuntu:~/lab03$ mkdir test
afzaalpc@ubuntu:~/lab03$ rmdir test
afzaalpc@ubuntu:~/lab03$ rm -r backup

```
afzaalpc@ubuntu:~/lab03$ touch test.txt
afzaalpc@ubuntu:~/lab03$ file test.txt
test.txt: empty
afzaalpc@ubuntu:~/lab03$ ls > output.txt
afzaalpc@ubuntu:~/lab03$ cat output.txt
file1.txt
output.txt
test.txt
afzaalpc@ubuntu:~/lab03$ ls >> output.txt
afzaalpc@ubuntu:~/lab03$ cat output.txt
file1.txt
output.txt
test.txt
file1.txt
output.txt
test.txt
afzaalpc@ubuntu:~/lab03$ ls | less
afzaalpc@ubuntu:~/lab03$ pwd
/home/afzaalpc/lab03
afzaalpc@ubuntu:~/lab03$ ls -la
total 16
drwxrwxr-x  2 afzaalpc afzaalpc 4096 Oct  8 04:32 .
drwxr-x--- 21 afzaalpc afzaalpc 4096 Oct  8 04:21 ..
-rw-rw-r--  1 afzaalpc afzaalpc   50 Oct  8 04:25 file1.txt
-rw-rw-r--  1 afzaalpc afzaalpc   60 Oct  8 04:32 output.txt
-rw-rw-r--  1 afzaalpc afzaalpc    0 Oct  8 04:30 test.txt
afzaalpc@ubuntu:~/lab03$ ^C
```

In